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B. E. (Sixth Semester) Examination, Nov.-Dec. 2015

(Old Course)

(IT Engg.)

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DATABASE MANAGEMENT SYSTEM

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Part (a) of each unit is compulsory & carry 2 marks. Attempt any two parts from (b), (c) and (d) & carry 7 marks from each unit.*

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Unit-I

1. (a) What are the four difference between a file processing system and DBMS?

(b) What is the purpose of generalization, specialization and aggregation? Explain with example.

(c) Construct a B+ tree for following set of key values :

5, 7, 11, 18, 20, 25, 28, 30, 35

Assuming that the tree is initially empty and values are added in the ascending order in a B+ tree of order 4. Show the form of the tree after each of the following series of operation :

(i) Insert 9

(ii) Delete 28

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(d) Construct an ER diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted?

Unit-II

2. (a) What do you mean by referential Integrity?

(b) Consider the employee database where the primary keys are underlined. Give expression in SQL for each of the following queries.

(i) employee (<u>employee name</u>, street, city)

(ii) works (<u>employee name</u>, company name, salary)

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(iii) company (company name, city)

(iv) manager (employee-name, manager name)

Queries are : CSVTUonline.com

(i) Find all employees in the database who do not work for 'ABC' company.

(ii) Find the names and cities of residence of all employees who work for 'ABC' company.

(iii) Modify the database so that 'Ravi' now lives in 'Bhopal'.

(iv) Give all employees of ABC company a 10% raise?

(v) Find name of all employees who work for 'ABC' company. CSVTUonline.com

(vi) Find all employees in the database. Who live in the same cities and on the same streets as do their manager.

(c) Explain the use of triggers for security purpose of database.

(d) Why we create views what is the difference between simple and complex view? Explain with some example.

Unit-III

3. (a) What is the need to normalize the logical design of database?

(b) Why BCNF is called strong form of 3 NF? Explain with example. CSVTUonline.com

(c) There is a relation schema R such that $R = (A, B, C, G, H, I)$ and functional dependencies are :

1. $A \rightarrow B$ 2. $A \rightarrow C$ 3. $CG \rightarrow H$ 4. $CG \rightarrow I$

5. $B \rightarrow H$

Find other FDs (F+). Also compute $(AG)^+$?

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(d) Suppose there in a relation $R = \{A, B, C, D\}$. The functional dependencies are :

1. $A \rightarrow BC$, 2. $B \rightarrow C$, 3. $A \rightarrow B$

4. $AB \rightarrow C$, 5. $AC \rightarrow D$

Find irreducible set of R?

Unit-IV

4. (a) What are the different phases of query processing?

(b) What are the different properties of transaction? Also explain different types of transaction failures.

(c) Explain in brief about serializability since every conflict serializable schedule is view serializable. Why do we emphasize conflict serializability rather than view serializability?

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(d) What is a recoverable schedule? Why is recoverability of schedule is desirable? Are there any circumstances under which if would be desirable to allow non-recoverable schedules? Explain your answer.

Unit-V

5. (a) What do you mean by Transaction Log?

(b) What do you mean by deferred update technique of recovery? Discuss its advantages and disadvantages.

(c) Explain Multiple granularity with an example.

(d) Write short notes on :

(i) Static two phase locking

(ii) Dynamic two phase locking

(iii) Strick two phase locking

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